

Tristan Hafer

Backend Developer

Orange County, CA • tristan@ultpcs.com • tristanhafer.com

SUMMARY

Backend-focused developer working primarily in Rust and PHP, building APIs, data pipelines, and performance-critical services for a medical device platform — from a production PHP API and MariaDB database to Rust services handling cellular eSIM management and WebAssembly-compiled data parsing libraries.

SKILLS

Languages: Rust, PHP, SQL, C/C++ **APIs & Services:** REST APIs (PHP), Rust HTTP clients (ureq), WebAssembly/FFI **Data:** MariaDB, SQL query authoring and analysis **Infrastructure:** Docker, self-hosted GitLab CI/CD

EXPERIENCE

Applied Cardiac Systems, Inc. — Irvine, CA *Project Manager, Software Developer* — September 2019 – Present

React Cloud Portal — API & data layer | 2020 – Present

- Built and maintain the PHP API backend powering the customer/internal study and monitor management portal
- Run and maintain SQL queries against the production `live_acsd_cloud` MariaDB database for monitor and eSIM data analysis

acsd-sdk / libwfdb — Rust/WASM ECG (WFDB) parsing library | Ongoing

- Built a Rust library for parsing ECG/WFDB physiological data, compiled to WebAssembly and published as a scoped npm package for frontend consumption
- Replaced panics (`unwrap()`/`expect()`) with proper `Result`-based error propagation
- Used `strum EnumDiscriminants` for FFI boundary mapping between Rust and JavaScript; debugged Cargo multi-target/wasm-pack conflicts and GitLab CI publishing pipeline issues

Teal eSIM Integration Tooling — Rust

- Built Rust tooling for Teal eSIM API integration, including deactivation workflows and clip-to-eSIM matching logic
- Used `chrono` for datetime handling and the `ureq` HTTP client for API communication

Study Archival Tool | 2024

- Built an automated Rust/Docker tool for archiving and packaging historical study records for seamless re-import

clip-file-daemon — Rust

- Built a Rust background service; resolved Docker networking issues and configured its GitLab CI/CD pipeline

Intranet — engineering support tooling | 2021 – Present

- Designed and deployed an internal Docker-based suite covering GitLab project management, CI/CD pipelines, VPN, and DNS

REFERENCES

Available upon request — including Wes Koerber (Sr. Software Developer) and Braden Walker (Sr. Full-Stack Web Developer)